



WOLLO University

Department of software engineering

Exit multiple choice question for

Software project management

Prepared by:

@Neyatibeb

KOMBOLCHA, Ethiopia

2015 E.C

Software Testing and Quality Assurance MCQ with Answers

1. When there are disagreements between the phase project manager and overall project manager, the matter should be escalated to the ____.

- a. Top-level Managers
- b. Upper Management
- c. Change Control Board
- d. CEO

Hide Answer

Answer: (C)

2. Which of the following is a snapshot of the project that gives a concise summary of the current condition of a project?

- a. Six Sigma
- b. Earned Value Analysis
- c. Software Metric
- d. Project Status Report

Hide Answer

Answer: (D)

3. The objective of ____ is to find problems and fix them to improve the quality of a project.

- a. Software bug
- b. Software complexity
- c. Software testing
- d. Software development

Hide Answer

Answer: (C)

4. Which type of testing process will create test scripts that will run automatically, repetitively, and through many iterations?

- a. White Box Testing
- b. Black Box Testing
- c. Manual Testing
- d. Automated Testing

Hide Answer

Answer: (D)

5. Unit testing is to test the ____ of the units.

- a. Performance

- b. System issues
- c. Functionality
- d. Hardware failure

Hide Answer

Answer: (C)

6. At the integration level, achieving the desired goal may be prevented in _____ testing if the test cases and results are not recorded properly.

- a. Bottom-Up
- b. Big Bang
- c. Top-Down
- d. System

Hide Answer

Answer: (B)

7. Which is an example of an indicator?

- a. Number of tests
- b. Number of staff-hours
- c. Actual versus planned task completions
- d. Defects per thousand lines of code

Hide Answer

Answer: (C)

8. Which of the following is used to collect direct measures of software engineering output and also its quality?

- a. Indirect measure
- b. Direct measure
- c. Function-oriented metrics
- d. Size-oriented metrics

Hide Answer

Answer: (D)

9. In _____, there are three characteristics that serve as a guide for the evaluation of a good design.

- a. Design and software quality
- b. Design concept
- c. Software design
- d. Modular design

Hide Answer

Answer: (A)

10. Which of the following defines the relationship between major structural elements of the software?

- a. Data design

- b. Linked list
 - c. Cohesion
 - d. Architectural design
- Hide Answer**

Answer: (D)

11. Cleanroom software engineering is an approach that emphasises the need to build _____ into software as it is being developed.
- a. Defects
 - b. Debugging
 - c. Correctness
 - d. Unit testing
- Hide Answer**

Answer: (C)

12. The projected usage of the software is analysed and a suite of test cases that exercise _____ of usage is planned and designed.
- a. An error record
 - b. A statistical quality control
 - c. Certification
 - d. Probability distribution
- Hide Answer**

Answer: (D)

13. Which of the following is a non-profit organisation and is also the world's leading professional association for the advancement of technology?
- a. IEEE
 - b. ANSI
 - c. CMMI
 - d. NDIA
- Hide Answer**

Answer: (A)

14. In which level of organisational maturity, conditions are not stable for the development of quality software?
- a. Optimising
 - b. Defined
 - c. Initial
 - d. Repeatable
- Hide Answer**

Answer: (C)

15. _____ products are capable of being used to generate entire applications from design specifications.

- a. CASE
- b. I-CASE
- c. Rapid Prototyping
- d. Repository

Hide Answer

Answer: (B)

16. The primary objective for tools in this category is to represent business data objects flow between different business areas within a company. Which category is this?

- a. Process Modeling & Management Tools
- b. Project Planning Tools
- c. Business Process Engineering Tools
- d. Risk Analysis Tools

Hide Answer

Answer: (C)

17. A _____ is a collection of objects or elements and is used as a cornerstone of formal methods.

- a. Set Operators
- b. Signature
- c. Set
- d. Union operator

Hide Answer

Answer: (C)

18. Which formal specification language can be used to describe the syntax of the programming language?

- a. Formal grammar
- b. Semantic domain
- c. Syntactic domain
- d. Sequence

Hide Answer

Answer: (A)

19. Software re-engineering can be defined as _____.

- a. The top-level process of engineering and a system to meet overall requirements.
- b. The examination and alteration of an existing subject system to reconstitute it in a new form.
- c. The engineering process of understanding, analyzing and abstracting the system to a new format at a higher abstraction level.
- d. The set of engineering activities that consumes the products and artifacts derived from legacy software and new requirements to produce a new target system.

Hide Answer

Answer: (B)

20. A maintenance organization's short-term goal is to clear the growing backlog of maintenance demands and the long-term goal is to support change at ____.

- a. Higher level
- b. Low level
- c. Requirements level
- d. Code-level

Hide Answer

Answer: (C)

21. Consider the below-mentioned statements:

- 1. A phased project manager monitors the overall project and is responsible to monitor the work of a project manager.
- 2. Phase project managers and overall project managers are together responsible for the contingency plans.

State True or False:

- a. 1-False, 2-False
- b. 1-True, 2-True
- c. 1-False, 2-True
- d. 1-True, 2-False

Hide Answer

Answer: (C)

22. Consider the below-mentioned statements:

- 1. White box testing technique guarantees that all independent paths within a module have been exercised at least once.
- 2. White box testing technique executes only one loop at their boundaries and within their operational bounds.

State True or False:

- a. 1-True, 2-False
- b. 1-False, 2-True
- c. 1-True, 2-True
- d. 1-False, 2-False

Hide Answer

Answer: (A)

23. Cutting out unnecessary requirements is called ____.

- a. Requirements Scrubbing
- b. Requirements Planning
- c. Requirements Scheduling
- d. Requirements Engineering

Hide Answer

Answer: (A)

24. Consider the below-mentioned statements:

- 1. Metrics do not establish a baseline from which improvements can be measured.
- 2. Metrics allow an organisation to identify the causes of defects which have the greatest effect on software development.

State True or False:

- a. 1-False, 2-False
- b. 1-True, 2-True
- c. 1-False, 2-True
- d. 1-True, 2-False

Hide Answer

Answer: (C)

25. Consider the below-mentioned statements:

- 1. Control hierarchy, also called a program structure, represents the organisation of program components (modules) and implies a hierarchy of control.
- 2. Control hierarchy represents procedural aspects of software such as the sequence of processes, occurrence or order of decisions or repetition of operations.

State True or False:

- a. 1-False, 2-False
- b. 1-True, 2-True
- c. 1-False, 2-True
- d. 1-True, 2-False

Hide Answer

Answer: (D)

26. Consider the below-mentioned statements:

- 1. In Cleanroom software engineering, the serious hazards can be related to human safety, economic loss or effective operation of the business and societal infrastructure.
- 2. The Cleanroom approach makes use of an outdated version of the incremental software model.

State True or False:

- a. 1-False, 2-False
- b. 1-True, 2-True
- c. 1-True, 2-False
- d. 1-False, 2-True

Hide Answer

Answer: (C)

27. Consider the following statements:

- 1. there is a strong correlation between ISO 9001 and the CMMI.
- 2. ISO 9001 addresses the minimum criteria for an acceptable quality system.

State True or False.

- a. 1- True, 2- False
- b. 1- False, 2- True
- c. 1- True, 2- True
- d. 1- False, 2- False

Hide Answer

Answer: (B)

28. An integrated CASE environment should provide the following:

1. Provide a mechanism for sharing software engineering information among all tools contained in the environment.
2. Enable a change to one item of information to be tracked to other related information items.

State True or False.

- a. 1- True, 2- False
- b. 1- True, 2- True
- c. 1- False, 2- False
- d. 1- False, 2- True

Hide Answer

Answer: (B)

29. Consider the following statements:

1. Thou shall not compromise thy quality standards: Expert training and ongoing consulting is essential for success when formal methods are used for the first time.
2. Thou shall document sufficiently: Formal methods provide a concise, unambiguous and consistent method for documenting system requirements.

State True or False:

- a. 1- True, 2- True
- b. 1- True, 2- False
- c. 1- False, 2- False
- d. 1- False, 2 – True

Hide Answer

Answer: (D)

30. Which of the following statements hold true:

1. Tools that support BPR include process modellers that allow organisations to run what-if scenarios on their key business processes.
2. BPR tools enables an organisation to set goals and gather information about defined and developed processes.

- a. 1-True, 2-True
- b. 1-False, 2- False
- c. 1- False, 2- True
- d. 1-True, 2-False

Hide Answer

Answer: (D)

31. Identify the correct statements regarding project metrics.

1. Project metrics and the indicators derived from them are used by a project manager and a software team to adapt project workflow and technical activities.

Department of software engineering

2. The first application of project metrics on most software projects occurs during the testing of the developed product.

3. Metrics collected from past projects are used as a basis from which effort and time estimates are made for current software work

4. Results of metrics can be used to provide an indication of the usefulness of work products as they flow from one framework activity to the next.

a. 1, 2 & 3

b. 1, 2 & 4

c. 2, 3 & 4

d. 1, 3 & 4

Hide Answer

Answer: (D)

32. For every software organization, the key element is ____.

a. People

b. Project

c. Process

d. Product

Hide Answer

Answer: (A)

33. SEI stands for ____.

A) System Engineering Institute

B) Software Engineering Institute

C) Software Engineers Institute

D) System Engineers Institute

Hide Answer

Answer: (B)

34. PM-CMM stands for ____.

A) Process Management Capability Maturity Model

B) Product Management Capability Maturity Model

C) People Management Capability Maturity Model

D) Project Management Capability Maturity Model

Hide Answer

Answer: (C)

35. PM-CMM was developed by ____.

A) IBM

B) IEEE

C) Microsoft

D) SEI

Hide Answer

Answer: (D)

36. _____ is responsible for total project management.

- A) Project Manager
- B) Project Developer
- C) Programmer
- D) System Manager

Hide Answer

Answer: (A)

37. PMI stands for _____.

- A) Process Management Institute
- B) Project Management Institute
- C) Project Mapping Institute
- D) Process Mapping Institute

Show Answer

38. DIN (Deutsches Institute for Normung) is a _____ organization.

- A) Process Management
- B) Product Management
- C) Standardization
- D) Software Development

Hide Answer

Answer: (C)

39. _____ consists of measuring and correcting activities to ensure that the goals are achieved.

- A) Staffing
- B) Quality Management
- C) Reporting Progress
- D) Controlling

Hide Answer

Answer: (D)

40. Analyzing progress compared to the baseline is known as _____ value management.

- A) Earned
- B) Spent
- C) Cost
- D) Time

Hide Answer

Answer: (A)

41. Creating a _____ is the first thing you need to do when undertaking any kind of project.

- A) Cost Estimation
- B) Project Plan
- C) Time Estimation
- D) Resources Estimation

Hide Answer

Answer: (B)

42. PERT stands for _____.

- A) Program Extraction and Review Technique
- B) Process Evaluation and Review Technique
- C) Program Evaluation and Reversing Technique
- D) Program Evaluation and Review Technique

Hide Answer

Answer: (D)

43. The bulk of the cost of software development is due to the _____ needed.

- A) Human Resources
- B) Software Resources
- C) Hardware Resources
- D) Machinery Resources

Hide Answer

Answer: (A)

44. Software Project Management begins with a set of activities that are collectively called _____.

- A) Cost Estimation
- B) Project Planning
- C) Time Estimation
- D) Resources Estimation

Hide Answer

Answer: (B)

45. The statement „Estimating is as much art as it is science“ is quoted by _____.

- A) Charles Babbage
- B) Pascal
- C) Frederick Brooks
- D) Von Neumann

Hide Answer

Answer: (C)

46. _____ model produces a software cost estimate as a function of a number of variables which relate to some software metric and cost drivers.

- A) Expert Judgment
- B) Analogy Estimation
- C) Top-Down Estimation
- D) Algorithmic

Hide Answer

Answer: (D)

47. A _____ consists of a list of a project's terminal elements with intended start and finish dates.

- A) Schedule
- B) Plan
- C) Prototype
- D) Estimation

Hide Answer

Answer: (A)

48. _____ can provide a graphical representation of a project schedule.

- A) Pie chart
- B) Gantt chart
- C) XY chart
- D) Bar chart

Hide Answer

Answer: (B)

49. The purpose of _____ is to plan how the activities in part or all of a project will be performed over a period of time.

- A) Analyzing
- B) Budgeting
- C) Scheduling
- D) Prototyping

Hide Answer

Answer: (C)

50. While scheduling, the activities to be performed are defined in _____.

- A) Project Plan
- B) Cost Plan
- C) Activity Plan
- D) Work Breakdown Structure

Hide Answer

Answer: (D)

Department of software engineering

51. _____ is an attempt to minimize the chances of failure caused by unplanned events.

- A) Risk Management
- B) Project Management
- C) Cost Management
- D) Quality Management

Hide Answer

Answer: (A)

52. Risk is the possibility of _____.

- A) Gain
- B) Loss
- C) Profit
- D) Credit

Hide Answer

Answer: (B)

53. There are _____ stages in the process of project risk management.

- A) Three
- B) Four
- C) Two
- D) Five

Hide Answer

Answer: (C)

54. _____ risks threaten the quality and timeliness of the software to be produced.

- A) Project
- B) Business
- C) System
- D) Technical

Hide Answer

Answer: (D)

55. _____ is an umbrella activity that is applied throughout the software process.

- A) Configuration Management
- B) Requirements Planning
- C) Requirements Scheduling
- D) Requirements Engineering

Hide Answer

Answer: (A)

56. _____ is a set of software engineering activities that occur after the software has been delivered to the customer.

- A) Analysis

- B) Support
 - C) Implementation
 - D) Testing
- Hide Answer**

Answer: (B)

57. _____ is a methodology to control and manage a software development project.
- A) Version Control
 - B) Change Control
 - C) SCM
 - D) Configuration Audit
- Hide Answer**

Answer: (C)

58. WBS stands for _____.
- A) Work Breakdown System
 - B) Work By Standard
 - C) Work Breakdown Structure
 - D) Work By System
- Hide Answer**

Answer: (C)

59. Consider the below-mentioned statements:
1. While conducting unit testing, the local data structure is examined to ensure that the temporarily stored data maintains its integrity during all the steps in an algorithm's execution.
 2. Black box testing is an effective technique for uncovering a broad array of path errors.
- State True or False:
- a. 1-False, 2-False
 - b. 1-True, 2-True
 - c. 1-True, 2-False
 - d. 1-False, 2-True
- Hide Answer**

Answer: (C)

60. _____ principle must be followed throughout the software development.
- A) Re-allotment
 - B) Incrementality
 - C) Decrementality
 - D) Reworking
- Hide Answer**

Answer: (B)

61. The aim of an organizational structure is to facilitate cooperation towards a common _____.
- A) Philosophy

- B) Business
 - C) Goal
 - D) Requirement
- Hide Answer**

Answer: (C)

62. The task of organizing can be viewed as building a _____.

- A) Project
 - B) Business
 - C) Process
 - D) Team
- Hide Answer**

Answer: (D)

63. _____ is an umbrella activity that is applied throughout the software process.

- A) Software Quality Assurance
 - B) Software Quality Management
 - C) Software Quality Testing
 - D) Software Quality Engineering
- Hide Answer**

Answer: (A)

64. The goal of software assurance is to reduce _____.

- A) Cost
 - B) Risks
 - C) Time
 - D) Quality
- Hide Answer**

Answer: (B)

65. FTR stands for _____.

- A) File Transfer
 - B) Formal Telephonic Review
 - C) Formal Technical Review
 - D) Formal Telegraphic Review
- Hide Answer**

Answer: (C)

66. _____ is a method used to identify defects in an artefact before progressing to the next stage of development.

- A) Testing
- B) Debugging
- C) Process

D) Formal Technical Review

Hide Answer

Answer: (D)

67. _____ analysis is a golden opportunity for process improvement that should not be missed.

- A) Project Closure
- B) Project Estimation
- C) Project Cost
- D) Project Schedule

Hide Answer

Answer: (A)

68. The data obtained during the closure analysis are used to populate the _____.

- A) Project Database
- B) Process Database (PDB)
- C) Database
- D) Records

Hide Answer

Answer: (B)

69. Many projects use the _____ method for estimation.

- A) Top-down
- B) Incremental
- C) Bottom-up
- D) Spiral

Hide Answer

Answer: (C)

70. The productivity of a project is measured in terms of _____ per person-month.

- A) Testing
- B) Debugging
- C) Codes Produced
- D) Function Points

Hide Answer

Answer: (D)

71. _____ provides a framework, from which, a comprehensive plan for software development can be established.

- A) Product
- B) Process

C) People
D) Project
Hide Answer

Answer: (B)

72. _____ is a collection of tasks handled in a planned and systematic order.
A) Plan
B) Product
C) Process
D) Project
Hide Answer

Answer: (D)

73. The factors influencing project management is/are _____.
A) Time
B) Cost
C) Scope
D) All of the above
Hide Answer

Answer: (D)

74. _____ stage determines the nature and scope of the development.
A) Planning
B) Design
C) Implementation
D) Initiation
Hide Answer

Answer: (D)

75. The items that comprise all information produced as part of the software process are collectively called a _____.
A) Software Process
B) Software Project
C) System Specification
D) Software Configuration
Hide Answer

Answer: (D)

